HIMANSHU SHUKLA

MACHINE LEARNING ENGINEERING | APPLIED AI

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PROFESSIONAL SUMMARY

Mathematics & Computing undergraduate building and testing applied machine learning systems. Project work spans satellite-image retrieval, contrastive embeddings, computer vision, vector search, model evaluation and Python APIs.

EDUCATION

Madhav Institute of Technology & Science (MITS), Gwalior | B.Tech, Mathematics & Computing
Expected graduation: 2028 | CGPA: 7.75/10 (through Semester IV)

TECHNICAL SKILLS

Machine learning: PyTorch, scikit-learn, contrastive learning, embeddings, FAISS, retrieval evaluation, F1@5/F1@10
Data and programming: Python, NumPy, Pandas, SQL, PostgreSQL, C++, MATLAB
Development tools: FastAPI, REST APIs, pytest, Docker, Alembic, Git, Jupyter Notebook
Additional exposure: React, TypeScript, Supabase, Scapy, webcam-based vision

PROJECT EXPERIENCE

EarthBridge | Cross-modal satellite-image retrieval
Python, PyTorch, FAISS, FastAPI | Research prototype

- Developed contrastive embedding retrieval for optical, SAR and multispectral satellite imagery; reported 90% retrieval accuracy in an internal project evaluation.
- Evaluated retrieval using leakage-aware splits, F1@5/F1@10 and latency checks; added a 128-pair overfitting check to inspect model behavior.
- Connected similarity search to a FastAPI demo and added pytest-based and artifact-integrity checks.

Mulyankan | Evidence-linked evaluation | 3D explanation
React, TypeScript, Supabase | Demonstrated prototype

- Developed document-to-rubric review workflows with evidence references, insufficient-evidence flags and human reviewer controls.
- Invited to demonstrate at CMPDI headquarters, Ranchi (11 July 2026); institute-level Smart India Hackathon selection.

IrisTrack | Webcam gaze and blink tracking | GitHub
Computer vision, webcam landmarks, calibration | Locally tested

- Implemented eye/facial landmarks, blink detection and gaze-pointer calibration; tested camera lifecycle and interaction flows on Windows.

Sentinel | Network monitoring and anomaly-analysis prototype
Python, Scapy, FastAPI | Local prototype

- Built packet/flow monitoring views and an initial anomaly-analysis workflow; verified website blocking through a local forward proxy.

ADDITIONAL PROJECTS

NewsSources (RSS/sitemap discovery, PostgreSQL and document-ingestion pipeline in development); OHS (React/Firebase real-time collaboration, video calling and synchronized Watch Together); Mulyankan 3D Spatial Lab (gesture-controlled architecture walkthrough).

LEADERSHIP & ACHIEVEMENTS

Class Representative (3rd Year); NPTEL Coordinator; Training & Placement Cell Member; SchoolBuddies co-founder. First Position, Neeti Summit English Debate; DevFusion 4.0 (advanced to Round 3); PyML - Python for Machine Learning.

RELEVANT COURSEWORK

Data Science, Linear Algebra, Numerical Techniques, Differential Equations, Data Structures and Algorithms, DBMS.